



The dynamic evolution mechanism of structural dependence characteristics in the global oil trade network

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ABSTRACT

This study aims to reveal the evolution mechanisms of the petroleum trade network by examining its dynamic evolution of structural dependence characteristics. We conduct an in-depth examination of the dynamic structural evolution of the global trade network by employing the TERGM and STERGM models, and utilizing oil trade data spanning the years 1997–2021. To understand the stability of oil network trade dependence, we incorporate the cascading failure and the ERGM models to simulate the impact of external supply crises on the structural dependency relations of the oil trade network. Our findings can be summarized as follows: first, reciprocal trade structures play a crucial role in building trade relationships, with countries emphasizing mutually beneficial partnerships. Second, with the increasing complexity of the global trade landscape, the stability and resilience of the oil trade network structure have experienced notable declines. Third, in response to external shocks, countries tend to diversify their oil trade relations to ensure energy supply security. The results suggest that the government should pay more attention to the oil trade dependency relationships.

1. Introduction

In the current global energy landscape, petroleum stands as the primary energy source, satisfying one-third of the world's energy demands [1]. According to Dudley [2], approximately 72.7 % of petroleum consumption relies on imports. The global pattern of oil trade and the intricate relationships among nations are diverse and complex [3]. Investigating the structural features of the global oil trade network and identifying its evolutionary mechanisms have become pivotal in comprehending the entire network [4,5], aiding in discerning significant influences on the network patterns ([6]; [7]). In recent years, several studies have focused on the structure of global trade networks to better understand the evolutionary characteristics of trade pattern networks ([8–14]). The evolution mechanism refers to the regulations governing internal organization and operational shifts within the network's structure [15]. Exploring the evolution mechanisms of networks can reveal the network's internal substructures and how changes in these structures affect the network. Investigating the evolutionary mechanisms of networks unveils the inherent substructures and elucidates the impact of alternations with these configurations on the network [16].

The interdependence between various parts of the network mechanism is crucial, as components within the network structure often form a more complex unit based on their interdependencies. Subsequently, these complex units influence the behavior of the mechanism [17]. Analyzing the evolving characteristics of internal dependency structures within the international petroleum trade network can provide deep insights into the evolution mechanisms of the petroleum trade network. However, these efforts have mostly concentrated on identifying inherent patterns within the oil trade network, paying limited attention to unveiling the mechanisms behind trade dependency on network patterns. Analyzing the internal dependency structure traits of the oil trade network facilitates the dissection of its evolutionary mechanism and identification of important influencing factors from a multifaceted standpoint, which can empower countries to leverage their respective strengths in addressing global geopolitical tensions and mitigating risks associated with energy collaborations. Thus, what constitutes the evolution mechanisms of the dependency structure of the global oil trade network? How does the network's dependency structure shift in the face of external shocks? Addressing these inquiries can provide profound insights into supply chains and spillover effects.

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Due to the geographical imbalance in global petroleum production and consumption, the global oil trade system is decentralized and heavily reliant on international trade, resulting in strong bilateral interdependencies. Hence, the intricate associations and mutually dependent structural patterns characterize the global oil trade. Structural dependency refers to the situation in which countries interact and establish network structures among each other, with network structures responding to interactions between countries [18]. Pan [19] noted that structural dependency might attenuate the significance of exogenous node attributes. Of particular note are the star, reciprocity, and triangle structures constituting the fundamental trade relationships within the global network, capturing trade dependency relations. However, structural dependence is inherently dynamic, with changes influenced by various interconnected social activities and processes. Specifically, network structures can transform by pursuing countries with elevated economic development or specific characteristics (such as resources and endowments) that favor reciprocal patterns. Moreover, trade network connections may arise from the network's inherent structural attributes, and relationships among trade participants can be influenced by other trade relationships within the same network. Additionally, uncertainties stemming from external factors such as global markets, policies, the COVID-19 epidemic, conflicts, and climate disasters can directly disrupt the structural dependency relationships within the oil trade, impacting the overall stability of the global oil trade. Consequently, delving into the dynamics and potential dependencies of the global oil trade offers a valuable and adaptable tool for establishing trade relationships.

This paper undertakes a comprehensive examination of the dynamic evolution mechanism of structural dependencies and potential dependence relationships within the complex network of the global oil trade. Specifically, we adopt the dynamic exponential random graph models (ERGMs) introduced by Hanneke et al. [20] and Krivitsky and Handcock [16] to quantify changes in dependency structures within the world oil trade. ERGMs additionally incorporate network dependency relationships, making the inclusion of dependency types more feasible [18]. ERGMs overcome the assumption of individual independence and emphasize the structural interdependencies between trade network structures. They include exogenous covariates and endogenous structural dependency features, to help explain the evolution of network structures by interpreting the significance of endogenous structural dependency variables at multiple micro levels. Notably, time-stamped random graph models (TERGM, the temporal exponential random graph model; STERGM, the separable temporal exponential random graph model), which are based on discrete-time network data, effectively utilize historical information to dynamically explore the correlation and topological dependency of network data across different periods. Moreover, by combining the cascading failure model with the ERGM, we simulate the impact of supply crisis shocks on the structural dependency relations of the global oil trade network.

Our research findings demonstrate that reciprocal trade relationships contribute to the establishment of trade ties, especially after financial crises, when countries are inclined toward reciprocal trade relationships. The negative periodicity and positive transitivity coefficients in the model indicate a hierarchical structural dependency within the oil trade network, which diminishes as the network scale expands. Once financial or supply crises occur, periodicity and transitivity exhibit growth trends, signifying shifts in oil trade patterns and structural dependency relations. The declining effects of eigenvector centrality, closeness centrality, and modularity further indicate the diminishing hierarchical effects within the oil trade network, reducing the degree of structural dependency. Additionally, since 2011, the structural stability of the oil trade has significantly decreased, aligning with the findings of Wei et al. [13], indicating substantial challenges facing the global oil trade. The dependency structure of the oil trade is susceptible to external influences. Countries lean toward diversifying oil trade relationships to ensure energy supply security.

This study makes several contributions. First, it reveals the dynamic

evolution mechanism of the internal dependence of the global oil trade network. In contrast to extant studies, we utilize ERGMs to investigate the characteristics of complex network dependencies in the global oil trade based on the perspective of oil trade structure dependence, and enhance the flexibility of incorporating dependency patterns into the model. For the first time, we employ the time variation dynamic models TERGM and STERGM, to identify the evolution of internal dependency features within the global oil trade and reveal their temporal trends. Second, the attribute variables of the model are expanded. We build on the original model by including centrality and modularity as actor attribute variables within the TERGM and STERGM models, further illustrating the trend of decreasing internal dependency within the oil trade network, an aspect not previously explored in the literature. Finally, we innovatively combine the cascading failure model with the ERGM to simulate the impact of an external supply crisis on the structural dependency relationships of the oil trade network. This innovative approach allows us to explore the internal features and changes in oil trade structural dependency, thereby holding significance for policy-makers and traders in enhancing the resilience of the global oil trade network.

The subsequent sections of this study are structured as follows: section 2 presents an overview of prior research on the complex network of the international oil trade, applications of the ERGMs, and avalanche models. Section 3 describes the data description and methodology, including definitions and explanations. Section 4 and section 5 provide the empirical results and corresponding discussion, respectively. Section 6 provides conclusions and policy implications.

2. Relevant literature review

Exploring the structural evolution mechanism of the global oil trade network can address the research gap in oil trade analysis related supply and demand using spatial economic models. Analyzing the structural characteristics of network enables us to comprehend complex correlation and interdependence patterns [21] and unveil the spatiotemporal evolution of energy resource flows [1,8,13,22–26] to gain insights into overall oil trade supply and demand patterns [27,28]. The complex network analysis approach for studying oil trade encompasses several facets. First, studies have employed topological features such as network density, clustering coefficients, and centrality to explore the overall trends in changes in oil trade relations and shifts in trade status rankings [29–31]. Second, complex networks can be integrated with other statistics models, such as the evolving nature of oil trade relationships [32–34]; [35]; [36].

Recent years, literature use multilayer network, gravity models and quadratic assignment models to analyze network mechanisms [10, 37–39]. These models often assume dyadic independence of trade relationships and overlook the internal variable selection, failing to consider the structural dependence relationship between countries. Structural dependence characteristics play a crucial role in obtaining a deeper understanding of the interconnectivity and coordination, as well as in analyzing the evolution mechanisms of the oil trade. However, the literature tends to focus more on identifying specific inherent patterns within the oil trade network, with limited analysis from the perspective of structural dependence.

To unveil the evolution mechanism of trade network relationships, scholars have amalgamated complex networks with ERGM models [40], [22,36,41–43]. ERGMs incorporate internal network relationships into structural statistics, elucidating the influence of early-stage relationship structures on subsequent formations [44], while accentuating the interdependence between network relationships [45]. For instance, Gao et al. [46] employed the ERGM to test key factors influencing technical cooperation among countries along the Belt and Road. However, the mechanisms governing the dynamic evolution of trade network structural dependencies, particularly in conjunction with time-series TERGM/STERGM models, remain a largely unexplored domain. The

TERGM and STERGM are frontier statistical models used to infer network formation mechanisms that effectively integrate assessments of endogenous dependence, exogenous country attributes, external influence effects, and temporal dependence effects. By employing these models, we can explain the evolution mechanisms of the oil trade network from the perspective of structural and temporal dependence characteristics by employing these models. Structural dependence refers to the connections between nodes in a network depending not only on whether they have direct links but also on the overall topology of the network. It does not obey the assumption of binary independence, but rather assumes that the topology of the network influences its evolution. For instance, nodes are more likely to establish connections with common neighbors or with nodes that have higher centrality. Our quantitative analysis indicates that the global oil trade network exhibits topological characteristics similar to those of real networks, such as “small-world” and “core-periphery structures”. The network shows distinct structural dependence features characterized by role-playing and mutual influence. Additionally, considering the long-term nature of cooperative relationships in countries’ oil trade, we also consider the possibility of dependencies between historical network structures and present network structures, namely temporal dependence. For example, countries that have engaged in petroleum trade in the past tend to become trade partners in the future. TERGM not only retains the flexibility of ERGM [20], but also accounts for time-varying factors, enabling the analysis of dynamic structure-dependent changes in trade relations. STERGM enhances historical network data utilization, augments model parameter interpretability, and exposes dynamic evolution mechanisms with trade networks by analyzing the formation of new relationships and the dissolution of original relationships [16,43,47]. For example, Matous et al. [48] employed TERGM to explore the impacts on low income communities. Cai et al. [49] and Guo et al. [50] have both utilized the TERGM to analyze distinct networks—the former focusing on the carbon emission correlation network and the latter on the Belt and Road natural gas trade network. Wang et al. [51] employed TERGM to dynamically analyze the establishment, continuation or disappearance of trade corporation relationships in the global strategic emerging industries. These studies have collectively affirmed the substantial influence of network structure on the creation and development of various networks. Therefore, building upon the static complex network model, this study innovatively integrates complex networks with the TERGM and STERGM models, empirically analyzes the evolution mechanism of the structural dependence characteristics of the global oil trade network from 1997 to 2021, and deeply reveals the intrinsic linkage of the evolution mechanism of trade relations.

To further assess the impact of supply crisis shocks on oil trade network structure dependence, we draw upon existing research concerning potential external risk shocks and the ripple effects of trade [52–55]. Building on this foundation, we attempt to evaluate the internal stability of oil trade network structure dependence through simulation. We employ a cascading failure model to simulate the propagation effects among key trading nations in the network, and then integrate the ERGM to comprehensively analyze how external shocks influence structural dependencies within the oil trade network, providing a more accurate reflection of its response to external disruptions.

In summary, the current literature predominantly examines binary relationships and fundamental topological structure indicators within trade relation networks. The use of constrained research methodologies limits the analysis of related trade networks from an exogenous variable perspective. The characteristics of structural dependence often go unnoticed, and the traits of trade relations are rarely illuminated from the viewpoint of dynamic evolution. Consequently, we investigate the global oil trade network through the lens of dynamic structural dependence and construct the TERGM and STERGM to analyze in depth the structural attributes of oil networks and the formation and evolution mechanism of state relations. Through the cascading failure model, we

test the resistance of the network’s structural dependence to external shocks, aiming to surpass the constraints of traditional binary relationship research, fill gaps in exploring the internal structure of trade relations, and furnish a scientific foundation for energy trade decision-making.

3. Methodology

3.1. Global oil trade network (GOTN) model

The global trade network can be depicted as a weighted network. Nevertheless, a weighted network can be simplified into a graph comprising nodes and edges, denoted as $G = (N, E)$. In the context of this study, the nodes correspond to countries, while the edges symbolize the connections within the oil trade relationships between these countries. The edge weight is represented by the trade volume, encompassing both the magnitude of trade and the direction of trade flow. We assume $V_i = [v_i] (i = 1, 2, \dots, n)$ to be export countries and $V_j = [v_j] (j = 1, 2, \dots, n)$ to be import countries. We define edges using the adjacency matrix $A_{ij} = [a_{ij}] (i = 1, 2, \dots, i; j = 1, 2, \dots, j)$, which characterizes the oil trade connection between countries i and j . We exclude self-loops. To provide a more accurate representation of trade relations, we selected oil trade values over \$10,000. Therefore, $G = (V_i, E_i, A)$ signifies the oil trade network under consideration.

3.2. Structural parameters of GOTN

(1) Degree

Degree denotes the count of edges linked to nodes i and j , serving as an indicator of a country’s extent of trade connections with other nations. This metric is further categorized into in and out degree, which respectively signify the capacity to initiate and receive trade relations. The calculation formula is provided in Equation (1), as follows:

$$ID_i = \sum_{j=1}^N a_{ji} \quad (1)$$

$$OD_i = \sum_{j=1}^N a_{ij} \quad (2)$$

where a_{ji}, a_{ij} indicate the elements within the directed adjacency matrix A .

Given their prevalence in real networks, the concept of weighted degree, signifying the summation of trade volumes across all edges, is commonly employed. Analogous to degree, weighted degree encompasses both weighted in and out degree. In the empirical analysis, the default calculation and visualization utilize weighted degree. The calculation formulas are shown in Equations (3) and (4).

$$S_i^{in} = \sum_{j=1}^N w_{ji} \quad (3)$$

$$S_i^{out} = \sum_{j=1}^N w_{ij} \quad (4)$$

where w_{ji}, w_{ij} represents the elements of the directed, weighted adjacency matrix W .

(2) Density

Network density measures the closeness of trade relations among countries in the network. The calculation formula is shown in Equation (5).

$$D = \frac{\sum_{ij} a_{ij}}{N(N-1)} \quad (5)$$

where D represents the network density, and $\sum_{ij} a_{ij}$ indicates the actual count of trade relationships within the network. N stands for the total of the countries.

(3) Reciprocity

Reciprocity is defined as the correlation coefficient between the adjacency matrix and represents the extent to which reciprocal trade relationships are established among countries in the network. The expression is shown in Equation (6).

$$R = \frac{\sum_{i \neq j} (a_{ij} - \bar{a})(a_{ji} - \bar{a})}{\sum_{i \neq j} (a_{ij} - \bar{a})^2} \quad (6)$$

where the average value $\bar{a} = \frac{\sum_{i \neq j} a_{ij}}{N(N-1)}$ measures the ratio of observed to potential links ($a_{ij} = 1$ if there exists a link from i to j , and $a_{ij} = 0$ if not). If $R > 0$, the oil trade network is reciprocal; if $R < 0$, it is a tri-ciprocal network.

(4) Clustering

The clustering coefficient assesses the level of clustering among nodes within the network. With n countries present, let's assume country i possesses k_i edges.

$$C_i^w = \frac{1}{s_i(k_i - 1)} \sum_{j,k} \frac{(w_{ij} + w_{ki})}{2} a_{ij} a_{jk} a_{ki} \quad (7)$$

$$C^w = \frac{1}{n} \sum_{i=1}^n C_i^w \quad (8)$$

C^w represents the average clustering coefficient across all countries. In the event of a relationship existing between countries i, j , and k to form a triangle, $a_{ij} a_{jk} a_{ki} = 1$. Conversely, if no such relationship exists, $a_{ij} a_{jk} a_{ki} = 0$.

(5) Betweenness Centrality

Betweenness centrality quantifies a country's role as an intermediary. If a country assumes a significant position between two other countries along their communication pathway, its betweenness centrality value is high. The formula is as follows:

$$B_i = \frac{1}{(N-1)(N-2)} \sum_{j < k} \frac{g_{jk}(i)}{g_{jk}} \quad (9)$$

where g_{jk} represents the count of shortest paths from country j to k , and $g_{jk}(i)$ indicates the number of shortest paths between nodes j and k that traverse node i . N stands for the total number of countries within the network.

(6) Eigenvector Centrality

The eigenvector centrality of a node typically significance of its neighboring nodes and suggests that the country is linked with numerous pivotal trade partners. The formula is provided in Eq. (10).

$$EC_i = \frac{1}{\lambda} \sum_{j \in N_i} A_{ij} EC_j \quad (10)$$

where N_i represents the set of nodes linked to node i , and A_{ij} is the adjacency matrix of the network. λ is a constant. When node j is connected to node i , $A_{ij} = 1$; or else, $A_{ij} = 0$.

3.3. ERGMs

The ERGM approach treats network as realization of random variables. Through statistical inference on a specific observed network, it becomes possible to estimate the attributes of the underlying distribution that result in the realization of the observed network. Building on the definitions provided by Lusher et al. [56] and Herman [18], the probability distribution function for the ERGM can be formulated in a general manner, as depicted in Equation (11).

$$Pr(Y=y|\theta) = \left(\frac{1}{c}\right) \exp\left(\sum_H \theta_H g_H(y)\right) \quad (11)$$

In the context, $c(\theta)$ serves as a normalization constant. Typically, y corresponds to the weight matrix A of the network. θ_H represents the attribute coefficient. H encapsulates the network attributes that influence the establishment of network relationships, delineated into three components: the endogenous structure effect of the network, the impact of actor relationships, and the external network effect. $g_H(y)$ denotes the sufficient statistics for the model, akin to a vector of clique potentials.

Assume a sequence of network sets $G_t = (V_t, E_t)$, where V_t signifies the nodes in the network at time t and E_t denotes the edges in the network at time t . A particular network configuration at time t is designated as y^t . According to the discrete-time Markov chain theorem, a k -order Markov correlation TERGM is introduced. In this context, the network structure in the t -th interval is solely dependent on the network patterns observed during the initial k intervals:

$$P(Y^t = y^t | Y^{t-k}, \dots, Y^{t-1}, \theta) = \frac{\exp(\sum_H \theta_H g(y^t, y^{t-1}, \dots, y^{t-k}))}{c(\theta, y^{t-k}, \dots, y^{t-1})}. \quad (12)$$

The TERGM can analyze network changes from the perspective of the entire historical period, but when analyzing network data, it is necessary to further consider the new generation of network relationships and the dissolution of original relationships [16]. Since there may be two different situations regarding dynamic change in the network, namely, the formation and dissolution of the relationship, the change in the network structure is also divided into two situations, namely, the establishment of a new relationship and the dissolution of the original relationship. The mechanisms behind these two changes are not the same. The TERGM does not consider these two situations separately, so Krivitsky and Handcock [16] proposed the STERGM, which decomposes the TERGM into two parts, the formation model and the dissolution model, and analyzes the generation and maintenance of network relationships.

Introduce the concept of the evolving network as $Y^+ = Y^{t-1} \cup Y^t$. If new relationships emerge during period t , Y^+ will encompass these relationships. Define the dissolution network as $Y^- = Y^{t-1} \cap Y^t$. If period t involves the release of relationships, then period t will exclusively include the remaining relations. the joint parameter vector of the formation model and dissolution model is denoted as $\theta = (\theta^+, \theta^-)$. The first-order Markov correlation STERGM is outlined as follows:

$$\begin{aligned} \llbracket P(Y) \rrbracket^t = y^t | Y^{t-1} = y^{t-1}, \theta &= P(Y^+ = y^+ | Y^{t-1} = y^{t-1}, \theta^+) \\ &\times P(Y^- = y^- | Y^{t-1} = y^{t-1}, \theta^-) \end{aligned}$$

Further transformed into :

$$P(Y^t = y^t | Y^{t-1} = y^{t-1}, \theta) = \frac{\exp(\sum_H \theta_H^+ g(y^+, y^{t-1}))}{c(\theta^+, y^{t-1})} \frac{\exp(\sum_H \theta_H^- g(y^-, y^{t-1}))}{c(\theta^-, y^{t-1})} \quad (13)$$

In this paper, the MCMC MLE method is used to estimate the coefficients of TERGM and STERGM.

3.3.1. Variables analysis

(1) Endogenous structure effect variables.

These coefficients discern the shifts trajectory of the network's dependency structure during the process of dynamic change. Positive and negative coefficients signify whether the likelihood of a structure manifesting in the network surpasses random expectations while other conditions remain unaltered. In this study, the number of edges, (delayed) reciprocity, transit closure and network stability are chosen as endogenous structure dependent variables. Reciprocity describes the tendency of countries to form mutually beneficial trade relations, and its positive effect makes the network tend to form more reciprocal structures, which in turn makes it easier to establish or maintain mutually beneficial trade relations between countries. Cyclicity and transitivity describe the tendency of countries in the network to trade in groups. The negative effect of cyclicity makes the loop structure in the network tend to decrease gradually, which makes it difficult to form or maintain the loop structure in the network. Transitivity describes the tendency of countries in the network to trade in groups. Stability describes the stability of the network pattern, and its positive effect makes the overall pattern of the network tend to be more stable. It is stable and less prone to major changes.

(2) The actor-relationship variable.

Some qualities or attributes of individuals in a network are crucial to the formation of network relationships, such as a country's factor endowment, economic development level, or other characteristics. This paper considers the structure dependent characteristic variables of the oil trade network as actor attribute variables to better explain the formation or maintenance of trade relations, and selects eigenvector centrality, betweenness centrality and modularity as the influence of countries in GOTN. A positive effect reflects that those countries have higher network influence, and it is easier to form or maintain trade relations between countries. We also utilize GDP as the representative of country development level and divide GDP into high-level and low-level countries to better discover the impact of GDP on dependence of oil trade structure.

(3) External network effect variable.

External network effect variables determine the degree of symbiosis between trade network relationships and other network relationships. The traditional gravity model believes that geographic distance will have a negative impact on the trade between two countries, so this paper uses the weighted network composed of geographic distances between countries as an exogenous network embedding model to test the degree of its impact. The negative effect of geographic distance reflects that the further apart the two countries are, the less likely they are to establish or maintain a trade relationship. In addition, language differences will also be detrimental to the formation of trade relations. In summary, we choose geographic distance and language as external network effect variables to add to the model.

3.4. Avalanche model

Dynamic shock propagation models are employed to elucidate the pathways of risk transmission and to identify pivotal risk sources within trade networks. Specifically, these models simulate the diffusion of shocks, particularly in scenarios of supply limitations, Xu et al. (2021) introduced a risk transmission model for trade networks under two supply shortage conditions, elucidating risk propagation within the EV-LIB trade network. We adopt a similar model to evaluate the risks stemming from supply restrictions in the global oil trade.

Within this network framework, nodes exhibit normal and avalanche

states, each possessing a certain resilience capacity against risks. Drawing from global oil trade network, we establish simulation parameters and distribution of avalanche sizes, formulated by the parameter ratio α/β . The dynamics of supply crisis propagation across the network, and consequently the distribution of avalanche sizes denoted as $P(A)$, hinge on the α/β parameter ratio [52,54,55]. For capturing tail characteristics, we select $\alpha/\beta = 1.5$ as a pivotal model parameter value, aligned with scenario analyses in the following section (see Appendix A.2). Subsequently, we compute both the avalanche size (A) and the avalanche duration (T), representing the number of steps in the crisis spread, for each country.

4. Empirical results

4.1. Data

We download bilateral trade flow data over 10 years, from 1996 to 2021, from the United Nations Commodity Trade Database (UN Comtrade; <http://comtrade.un.org>). The specific code used is HS 270900, which corresponds to crude petroleum oils. To represent the current developmental levels of countries, we incorporated GDP data sourced from the World Bank. Geographical distance and language data were sourced from the CEPII database.

The historical oil import and export trade values from 1996 to 2021 are illustrated in Fig. 1. Notably, in the period to 2008, the oil import and export trade volumes exhibited a rapid increases. The global economic crisis in 2008 precipitated a steep decline in oil trade volume, which reached a noteworthy peak from 2011 to 2014. Following this, advancements in oil extraction technology, such as Canadian oil sands and US shale oil, coupled with the fundamental impact of global financial markets, led to a marked reduction in oil prices and, consequently, a significant decrease in import and export trade volume. The emergence of the COVID-19 pandemic in 2020 and the concurrent global trade tensions contributed to a surge in oil prices, albeit accompanied by a reduction in oil exports.

4.2. Evolution of the global oil trade network

We use oil trade data from 1996 to 2021 to construct a statically weighted directed trade network. Each node represents a country, and the edge symbolizes trade interactions between two countries. Arrows are oriented from the exporting country to the importing country, and the magnitude of trade between nations is reflected by the thickness of the connecting lines, thereby representing the weight within the network. Additionally, the dimensions of nodes and labels are determined by the node weights.

Given the considerable number of network diagrams involved, we chose to focus on four representative years, 1998, 2010, 2015, and 2021, to visually depict the evolution of the trade network, as shown in Fig. 2. The USA and RUS have consistently held core positions within the global petroleum trade network. As China's oil trade relationships diversify, its centrality is gradually increasing. Against the backdrop of the rapid advancement of economic globalization, the status of oil trade within the global trade arena has gained prominence, accompanied by an increase in its market influence. Presently, fluctuations in oil prices have emerged as a pivotal for measuring the stability of worldwide trade. From the perspective of the static oil trade network diagrams, it is evident that prior to the onset of the pandemic, the structure of the oil trade exhibited annual growth in complexity. An expanding number of countries actively participated in the oil trade network, leading to a gradual erosion of the delineations between oil trade associations. Consequently, the trade relationships between countries ceased to be simplistic and single-dimensional. In summary, the oil trade network has exhibited expansion, alongside an escalation in the intricacy of its trade structure.

We employ Gephi software to calculate the pertinent metrics of the

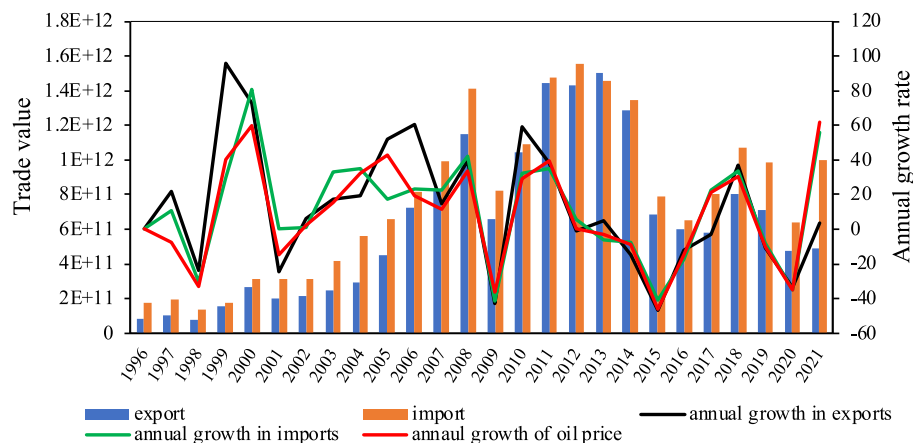


Fig. 1. Trade value from 1996 to 2021 (unit: US\$ billion). The y-axis on the left represents the trade value and on the right represents the annual growth rate, and the x-axis represents the year. In the bar graph, blue represents the export and red is the import. In the line chart, black, green and red represent the annual growth rate of exports, the annual growth of imports and the annual change of oil prices, respectively.

Global Oil Trade Network (GOTN) from 1996 to 2021, as displayed in Table 1. This analysis allowed us to scrutinize the evolutionary traits of structural dependence within the world oil trade network on a comprehensive level.

Over the years, both the nodes and edges of the GOTN displayed an upward trajectory until 2018, signifying the expanding scope of the oil trade network. Conversely, from 2019 onward, the connectivity and size of the GOTN have notably decreased. The surge in participant countries has led to an approximately 14 % reduction in its average path length. The gradual increase in the overall network density of the global oil trade mirrors the growing interconnectedness among nodes each year. Similarly, the average clustering coefficient, also referred to as transitivity, follows a parallel trend with network density, indicating an escalating level of oil trade clustering until 2019.

The modularity, which gauges the degree of compartmentalization within the network, experienced a substantial decline from 2014 to 2017. This observation suggests that trading countries were forming closer ties within the oil trade network during that period. Conversely, starting in 2019, the modularity displayed an upward trajectory, signifying a partitioning of the dependency structure within the GOTN. In a holistic context, propelled by global economic globalization, the GOTN has been gradually expanding while the depth of its structural dependence has been increasing. However, owing to the profound impact of the COVID-19 pandemic and the uncertain landscape of worldwide economic policies, the pattern of oil trade has reversed, disrupting the trend of multilateral trade and concurrently weakening the structural dependence.

4.3. Analysis of the evolution mechanism of the structural dependence of GOTN

The ERGM can effectively explain the network formation mechanism observed at a certain point in time, but it is not suitable for analyzing the dynamic changes in trade network structural dependence. The TERGM and STERGM analyze the multiyear trade network as a whole and fully consider the impact of historical network patterns. Therefore, we mainly apply these two models to analyze the mechanisms underlying the dynamics of global oil trade networks.

4.3.1. ERGM and TERGM tests

To verify the fitting effect of the TERGM, we employ two distinct methodologies, namely the ERGM and TERGM, to fit the network data series spanning from 2011 to 2020. The outcomes of these analyses are presented in Table 2.

From the insights shown in Tables 2 and it is evident that the

estimated coefficients for the same attribute variable differ significantly between the ERGM and TERGM models. The edge coefficient increases from -4.089 to -1.499 , the positive effect of reciprocity becomes insignificant, and the cyclicity coefficient decreases to a very significant level, indicating that the dynamic exponential random graph model has a significant impact on the formation of network relationships and can better reflect the revolution mechanism of the network structure.

We next compare the feature indicator values of the simulated network graph with the actual observed network feature indicator values. A box plot is drawn by simulating the values of the network characteristic indicators; the closer the midpoint is to the actual observed characteristic index value point, the better the model fitting effect [42]. Fig. 3 shows the statistical results of simulating 100 networks according to the estimated parameters of the two models, ERGM (a) and TERGM (b), in the regression results of Table 2. The initial five subgraphs present a juxtaposition between the simulated and observed values of the receiver operating characteristic (ROC) curve, marked by red line. A ROC curve positioned closer to the upper left corner is indicative of a more favorable model performance. Notably, the TERGM exhibits a superior fitting effect, as evidenced by the proximity of its ROC curve to the desired position.

4.3.2. Identifying the evolution mechanism of oil trade patterns and dependencies via the TERGM and STERGM

Based on the above analysis, we utilize the TERGM and STERGM to delve into the mechanisms driving the dynamic changes in the global oil trade network. Due to limitations imposed by the model, which ask for the common country nodes in the period, we exclude samples from the year 1996. We use oil trade data spanning from 1997 to 2021. The outcomes of our models are illustrated in Table 3.

Evaluation of the coefficient estimation results for network endogenous structural variables indicates distinct trends. The reciprocity coefficient, while nonsignificant in TERGM, is significantly positive in the formation results of STERGM but negative in the dissolution context, albeit without statistical significance. This pattern suggests a noteworthy impact exerted by one country on another. The establishment of an initial one-way trade relationship appears conducive to subsequent mutual trade ties, contributing to trade stability. In TERGM, the cyclicity coefficient is significantly negative at the 1 % level, while the transitivity coefficient is significantly positive. This underscores the presence of a pronounced hierarchical effect within the oil trade network, wherein trade relationships gravitate toward fixed directions. These hierarchical tendencies are reinforced by the STERGM findings. The coefficient of cyclicity displays significant negativity in information outcomes, but shows significance in a positive direction in dissolution results,

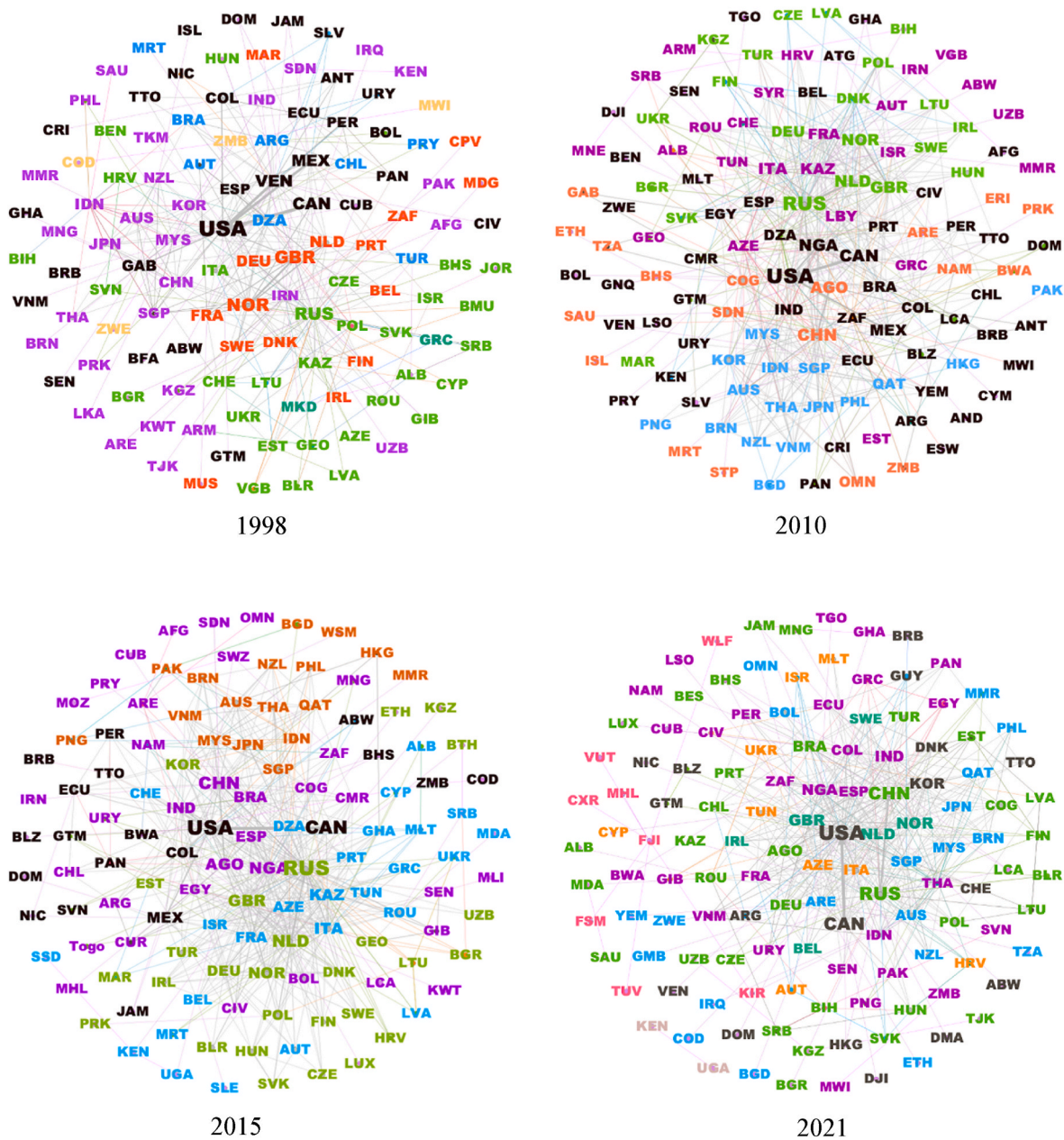


Fig. 2. illustrates the Global Oil Trading Network for the years 1998, 2010, 2015, 2021. Each individual point on the graph corresponds to a country participating in the oil trade. The utilization of distinct colors indicates different communities within the network. The size of each points is indicative of the level of influence a particular country wields within the network. Additionally, the width of each edge signifies the wight of oil trade, while the direction of an edge illustrates the direction of the oil trade flow.

Table 1
Descriptive statistics for the evolution of GOTN topology over time from 1996 to 2021.

Variable	Mean	Median	Sd	Min	Max	Range
Node	136.6	136	18.39	77	162	85
Edge	831.5	610	464.3	224	1548	1324
Density	0.036	0.033	0.008	0.026	0.059	0.033
Diameter	8	8	0.894	6	10	4
Average clustering	0.209	0.201	0.042	0.150	0.347	0.197
Average path	3.140	3.209	0.308	2.451	3.997	1.546
Modularity	0.442	0.446	0.069	0.305	0.570	0.265

indicating its inhibitory effect on new trade relation formation. Transitivity, with a coefficient of 0.90 in the formation result and -0.63 in the dissolution result, both of which are significant at the 1 % level, promotes new trade relationships and impedes trade dissolution-a congruence with established literature. The stability coefficient in the TERGM stands at 1.816, indicating 1 % significance and implying sustained stability of the global oil trade network. This coherence endures despite structural transformations that have occurred since 1997.

Turning to the actor relationship effect variables, the coefficient of GDP remains nonsignificant in TERGM. However, in the formation and dissolution models under the STERGM, the homophily coefficients for countries with low GDP attain significance, suggesting stronger tendencies for these countries to form or maintain trade relations. Betweenness centrality, reflecting intermediary status, and eigenvector centrality, denoting influence within the network, both yield

Table 2

Comparison of ERGM and TERGM results. The network structural dependence result of ERGM is 2020. The result of TERGM is from 2011 to 2020.

Network Statistics	ERGM	TERGM
Edges	−4.08827*** (0.194)	−1.49848 *** (0.143)
Reciprocity	0.74096 ^b (0.242)	−0.0960 (0.112)
Cyclicity	−0.02654 (0.020)	−0.14232*** (0.0046)
Transitivity	1.15936*** (0.158)	0.76365*** (0.038)
Homophily (High GDP)	0.19242 (0.098)	−0.03585 (0.048)
Homophily (low GDP)	−0.58784 (0.369)	−0.08316 (0.085)
PageRank	0.12588 ^a (0.026)	0.00137 (0.002)
Eigen centrality	−0.04593 ^c (0.033)	0.06409*** (0.019)
Betweenness centrality	0.13920 *** (0.030)	0.22531*** (0.018)
Modularity	0.65814*** (0.105)	0.52860*** (0.047)
Edg cov (Col net)	0.53717*** (0.124)	0.18742* (0.072)
Edg cov (Distance net)	0.09089 (0.063)	−0.40806*** (0.137)
Stability		1.61667*** (0.024)

Note: Standard deviation in parentheses. Signif. codes: 0.

^a 0.001.

^b 0.01.

^c 0.05 ‘.’ 0.1 ‘‘ 1.

significantly positive coefficients on formation outcomes for both TERGM and STERGM, and significantly negative coefficients on dissolution, accentuating their role in fostering network relationships. The modularity coefficient has a significantly positive value, validating the hierarchical network structure. Notably, the coefficients for these variables are consistently positive for formation in both TERGM and STERGM but significantly negative for dissolution, signifying their positive influence on oil trade formation between nations.

In terms of external influence factors, the net coefficient of the variable Col exhibits strong significance (at the 1 % level) for TERGM and STERGM formation, indicating that shared language environment facilitates oil trade relationship formation. Moreover, the net coefficient for distance is significantly negative (at the 5 % level) in the TERGM, indicating that geographical distance has a detrimental impact on the establishment and maintenance of oil trade relations and reaffirming the conclusions drawn from traditional gravity models.

Moreover, upon analyzing the trend chart of the oil trade volume and network structure, a notable impact of the global economic crisis in 2008 on the oil trade becomes apparent. To assess whether this financial crisis led to structural shifts in the global oil trade, we partitioned the model data into precrisis and postcrisis periods for analytical purposes.

Upon comparing the coefficients of the endogenous structural variables in Tables 4 and 5, a striking evolution is observed. The reciprocity coefficient in TERGM lost significance after the 2008 crisis. Conversely, under STERGM formation, this coefficient increases from 0.417 to 1.864, while under dissolution, it decreases from 0.386 to −0.084. This indicates a noteworthy surge in the positive impact of reciprocity on the establishment of trade relations postcrisis. Reciprocity plays an increasingly vital role in forming trade relations, aligning with the theory of reciprocity preference in economics and governments’ heightened focus on reciprocal trade protection. Regarding cyclicity, in both the TERGM and STERGM formations, its significantly negative coefficient remains consistent before and after the crisis. This suggests a declining cyclic structure in the network and an inclination of countries within the oil trade network to forge more relations, reflecting an ongoing expansion of the trade network’s complexity in trade relations.

The stability coefficient of the trade structure in TERGM decreased from 2.16 to −0.41, and both are statistically significant. This suggests a marked reduction in the stability of the oil trade network structure from 2011 to 2020. Amid the accelerating global energy transition and the evolving complexities within the international crude oil market, structural instability arises from factors such as U.S. shale oil, geopolitical conflicts, new energy sources, extreme weather events, and trade disputes.

For the actor relationship effect variables, the assortative coefficients

of national GDP levels remain nonsignificant in the TERGM results both before and after the financial crisis. However, in the formation model under the STERGM, from 1997 to 2008, high GDP levels yield significantly negative incompatibility coefficients, while low GDP levels yield significantly positive coefficients. This indicates that GDP does not foster oil trade relations, substantiating the hierarchical and community structure of the oil trade network. Postcrisis, the impact of national GDP levels on oil trade weakens.

Following the crisis, the eigen centrality and betweenness centrality coefficients significantly decrease for both the TERGM and STERGM formations, but increase for dissolution. The U.S.’s shale technology revolution, coupled with production surges in Canada and the Western Hemisphere, has reconfigured global oil supply and demand patterns, minimizing the importance of network centrality. Modularity follows a similar trend, weakening the role of the hierarchical structure.

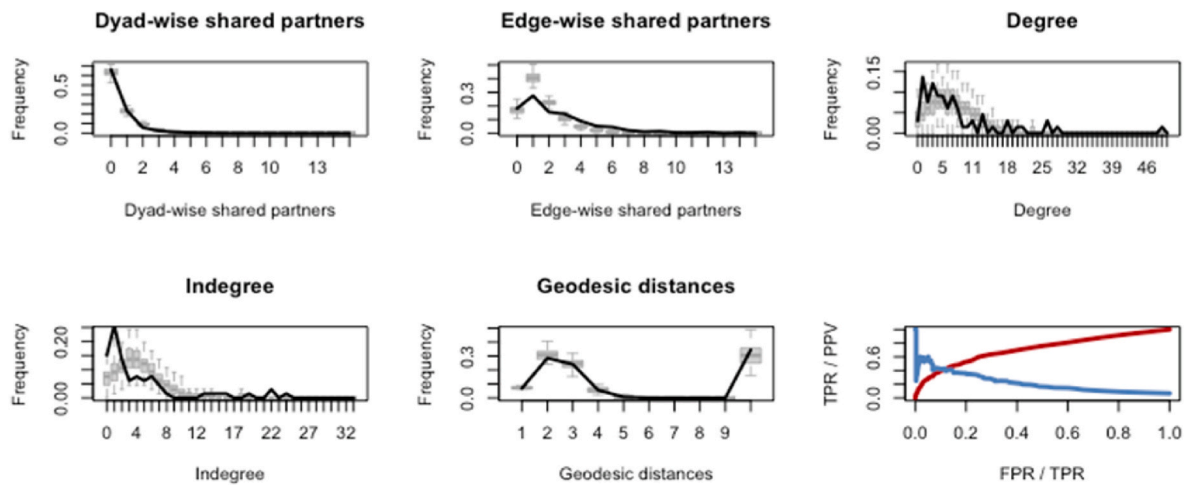
Since 2011, language has had significantly influenced the oil trade. In the expanding oil trade network, countries sharing a language are more inclined to establish new trade relationships.

In summary, reciprocity emerges as a critical factor in fostering oil trade relations between countries, serving as a foundation for the establishment of trade connections. Mutual trade relationships assist governments in overcoming the prisoner’s dilemma inherent in terms of trade, ultimately leading to benefits for all parties involved. With ongoing advancements in science and technology, oil has gained prominence as a pivotal energy source driving the development of nations. The global oil trade network has correspondingly expanded, particularly in the post financial crisis era. The influence of endogenous structural features within the oil trade network has gradually waned over time. The effects of variables such as GDP, eigenvector centrality, and intermediary centrality have displayed a declining trend. This can be attributed to the evolving nature of the global energy landscape and the intricate interplay of economic and geopolitical factors. Despite the expansion of the oil trade network and the escalating complexity of its structure, a discernible decline in the stability of this structure has transpired, accompanied by heightened trade uncertainty. As the world navigates challenges ranging from energy transitions to geopolitical dynamics, the dynamics of the oil trade network continue to evolve. Understanding these intricate relationships and their underlying mechanisms is paramount for governments, policymakers, and stakeholders seeking to navigate the complexities of the global oil trade.

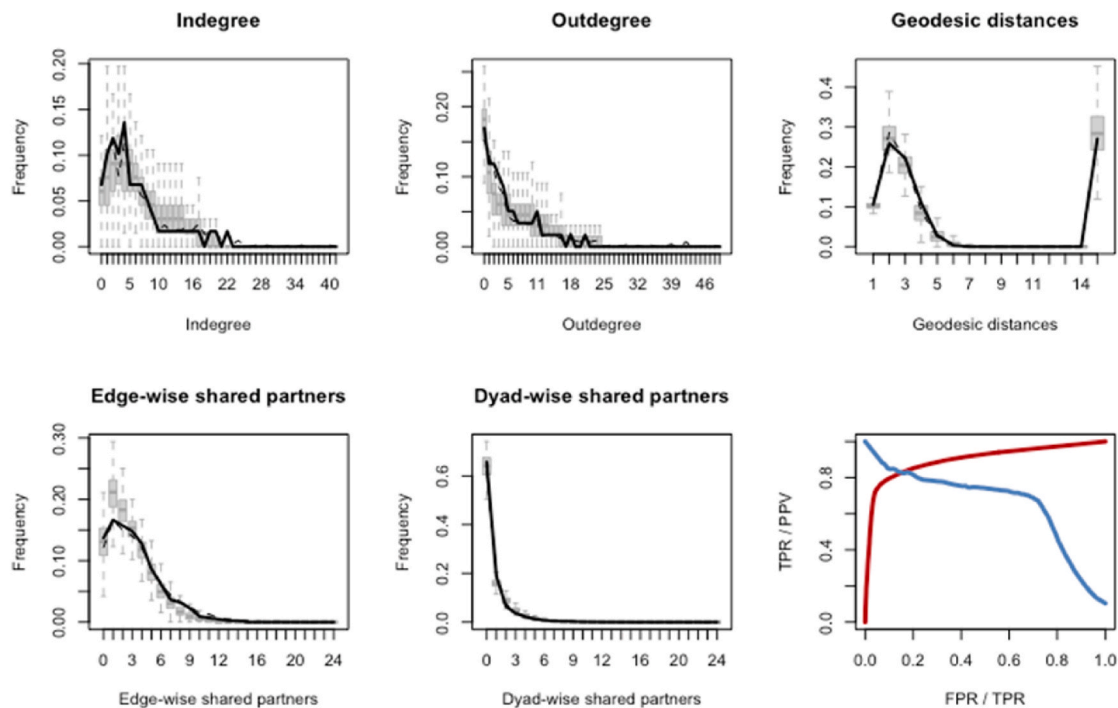
4.4. The influence of external shocks on the structural interdependence of the global oil network trade

The empirical analysis results from both the TERGM and STERGM highlight a significant decrease in the stability of the oil trade network structure, attributed to the amplification of globalization and the escalation of external shock factors. The contemporary demand for oil is closely tied to economic trends, and geopolitical considerations have emerged as pivotal disruptors in the realm of oil trade. As the world grapples with the ongoing repercussions of the COVID-19 pandemic and a broader economic downturn, disruptions within the oil supply chain could trigger shifts in trade patterns and exacerbate the instability of trade structures. To comprehensively assess these dynamics, we employ a cascading failure model coupled with the ERGM to simulate the effects of a global oil supply crisis on the oil trade network. This enables us to delve deeper into the network’s structural stability and internal dependence characteristics.

Our approach commences by calculating the distribution of avalanche sizes and plotting the complementary distribution on a log-arithmetic scale, following the methodology outlined by Sun et al. [54]. Although the intricate details are not displayed here due to space constraints, this approach effectively captures the tail characteristics of the distribution. Subsequently, we construct an avalanche network, as depicted in Fig. 4, which serves as a visual representation of the negative impact on the global oil trade network. Notably, the avalanche effects



(a) ERGM model fitting results



(b) TERGM model fitting results

Fig. 3. ERGM (a) and TERGM (b) model fitting results. A closer alignment of the midpoints with the actual observed characteristic index values indicates improved model fitting efficiency. Similarly, a ROC curve positioned nearer to the upper left corner signifies enhanced model performance.

are most pronounced for the USA, NGA and RUS. These countries experience the most substantial disruptions, leading to a discernible weakening of structural dependence.

Within this network, three prominent communities emerge, with the USA serving as a central hub in the oil trade landscape. Its influence spans across various regions and is particularly far reaching. In the event of a supply crisis, the ramifications reverberate directly across Europe and some Asian nations. Similarly, the impact of the ROU primarily extends to European and select Asian countries, while the NGA

predominantly affects countries within the Asian region. On the whole, the network's structural dependence remains robust, yet the potential impact of an external shock has the capacity to reshape the dynamics of structural interdependencies.

To further investigate the evolutionary trends of structural dependence characteristics, we employ the ERGM based on the avalanche network. The outcomes are presented in Table 6. We compare the ERGM results under three scenarios: normal, small, and large oil supply crises, aiming to emphasize the significance of intricate oil trade patterns in

Table 3

TERGM and STERGM results from 1997 to 2021.

Network Statistics	TERGM	STERGM	
		Formation	Dissolution
Edges	−1.40651 *** (0.109)	−4.63770*** (0.043)	0.22871 *** (0.055)
Reciprocity	−0.12681 (0.104)	1.46352*** (0.063)	−0.08858 (0.096)
Cyclicity	−0.15629*** (0.004)	−0.35527*** (0.019)	0.39087*** (0.031)
Transitivity	0.76365*** (0.038)	0.90103 ^a (0.034)	−0.62923*** (0.039)
Homophily (High GDP)	−0.02479 (0.038)	0.00231 (0.045)	0.08143 (0.060)
Homophily (low GDP)	−0.10652 (0.070)	0.32491** (0.125)	0.62628*** (0.172)
Eigen centrality	0.09482*** (0.014)	0.14129*** (0.017)	−0.11931*** (0.021)
Betweenness centrality	0.20257*** (0.013)	0.19653 ^a (0.012)	−0.12052*** (0.017)
Modularity	0.67693*** (0.036)	0.79664*** (0.041)	−0.90053*** (0.052)
Edgecov (Col net)	0.17218*** (0.050)	0.27544 ^a (0.053)	−0.12733 (0.070)
Edgecov (Distance net)	−0.28784 ^b (0.109)	0.02113 (0.054)	−0.00879 (0.027)
Stability	1.81645*** (0.018)		

Note: Standard deviation in parentheses. Signif. codes: 0 *0.05 ‘.’ 0.1 ‘.’ 1.

^a 0.001.^b 0.01.**Table 4**

1997–2008 TERGM and STERGM model results.

Network Statistics	TERGM	STERGM	
		Formation	Dissolution
Edges	−1.423859*** (0.190)	−5.10995*** (0.070)	−0.17443* (0.086)
Reciprocity	0.522423 ^c (0.231)	0.41670** (0.133)	0.38557* (0.185)
Cyclicity	−0.18941*** (0.009)	−0.32381*** (0.017)	0.33032*** (0.062)
Transitivity	0.74671*** (0.055)	1.02782*** (0.051)	−0.60023*** (0.060)
Homophily (HighGDP)	−0.04090 (0.071)	−0.36582*** (0.036)	0.30951** (0.106)
Homophily (low GDP)	−0.1338 (0.107)	0.33734*** (0.124)	0.13663 (0.145)
Eigen centrality	0.16150 *** (0.027)	0.33733*** (0.028)	−0.18614*** (0.037)
Betweenness centrality	0.17927*** (0.024)	0.17178 ^a (0.022)	−0.02168*** (0.029)
Modularity	0.81320 ^a (0.061)	1.29196*** (0.073)	−0.99031*** (0.091)
Edgecov (Col net)	0.15415 (0.087)	0.26556 ^b (0.099)	−0.10259 (0.125)
Edgecov (Distance net)	−0.0490 (0.189)	0.03172 (0.036)	−0.04571 (0.047)
Stability	2.15522*** (0.033)		

Note: Standard deviation in parentheses. Signif. codes: 0.

^a 0.001.^b 0.01.^c 0.05 ‘.’ 0.1 ‘.’ 1.

discerning trade relationships. The normal oil trade situation involves estimates utilizing the actual global oil export trade networks of 2021. On the other hand, small and large oil supply crises correspond to avalanche networks triggered by different degrees of shocks, aligning with avalanche parameters $\alpha/\beta = 1$ and $\alpha/\beta = 1.5$, respectively. The edge term functions as a constant, determining the baseline likelihood of

Table 5

2011–2021 TERGM and STERGM model results.

Network Statistics	TERGM	STERGM	
		Formation	Dissolution
Edges	−1.49848 *** (0.143)	−4.28258*** (0.069)	0.56568*** (0.075)
Reciprocity	−0.0960 (0.112)	1.86391*** (0.074)	−0.08412 (0.121)
Cyclicity	−0.14232*** (0.0046)	−0.35322*** (0.022)	0.37248*** (0.040)
Transitivity	0.76365*** (0.038)	0.76858*** (0.058)	−0.67322*** (0.056)
Homophily (High GDP)	−0.03585 (0.048)	0.06362 (0.057)	−0.06266 (0.077)
Homophily (low GDP)	−0.08316 (0.085)	0.01586 (0.111)	−0.44738*** (0.118)
Eigen centrality	0.00137 (0.002)	0.06545 ^b (0.021)	−0.09908*** (0.027)
Betweenness centrality	0.06409*** (0.019)	0.22134*** (0.016)	−0.17661*** (0.024)
Modularity	0.22531*** (0.018)	0.59936*** (0.057)	−0.76222*** (0.070)
Edgecov (Col net)	0.52860 ^a (0.047)	0.16029*** (0.073)	−0.13185*** (0.087)
Edgecov (Distance net)	0.18742 ^c (0.072)	0.00741*** (0.027)	0.00692 (0.034)
Stability	−0.40806*** (0.137)		

Note: Standard deviation in parentheses. Signif. codes: 0.

^a 0.001.^b 0.01.^c 0.05 ‘.’ 0.1 ‘.’ 1.

trade link formation. Negative values of the edges term could signify baseline trade costs. Regarding the reciprocal term, with the occurrence and expansion of the supply crisis, its value and significance gradually diminish. This implies that the trend of mutual trade relations is declining, indicating that countries face challenges in sustaining bilateral trade ties during external shocks. The negative coefficient of cyclicity suggests its unfavorable impact on the formation or maintenance of structurally dependent trade relations. The cyclicity coefficient shows a rising trend, and increases from −0.78 to −0.41. This suggests that the occurrence of an oil supply crises will break the original inherent trade structure and cause the country to seek new trade relationships. The transitivity term has the same change trend as cyclicity. Its coefficient is positive and gradually increases, indicating that the positive effect is increasing and that countries tend to build more shared trading partners to expand trade relations.

Overall, when encountering a large external shock, the structural dependence of the GOTN will change adaptively. Countries are no longer limited to establishing bilateral mutually beneficial trade relations but tend to break through the original trade hierarchy and actively explore more diversified trade relations.

4.5. Placebo test

To validate and ensure the stability of the model, this study construct models based on different time windows, including two years, three years and five years. The outcomes from these alternative time windows closely resemble the results obtained from the one-year time window, affirming the robustness of the model.

Additionally, the robustness of the coefficient estimates for TERGM and STERGM is assessed using both the bootstrap maximum likelihood estimation (MLE) method and the Markov Chain Carlo (MCMC) method, as utilized by Hunter and Handcock [57]. The findings indicate that the coefficients' signs remain consistent and closely approximate the MLE estimates. Consequently, the results obtained from the ERGM, TERGM and STERGM models demonstrate strong robustness.

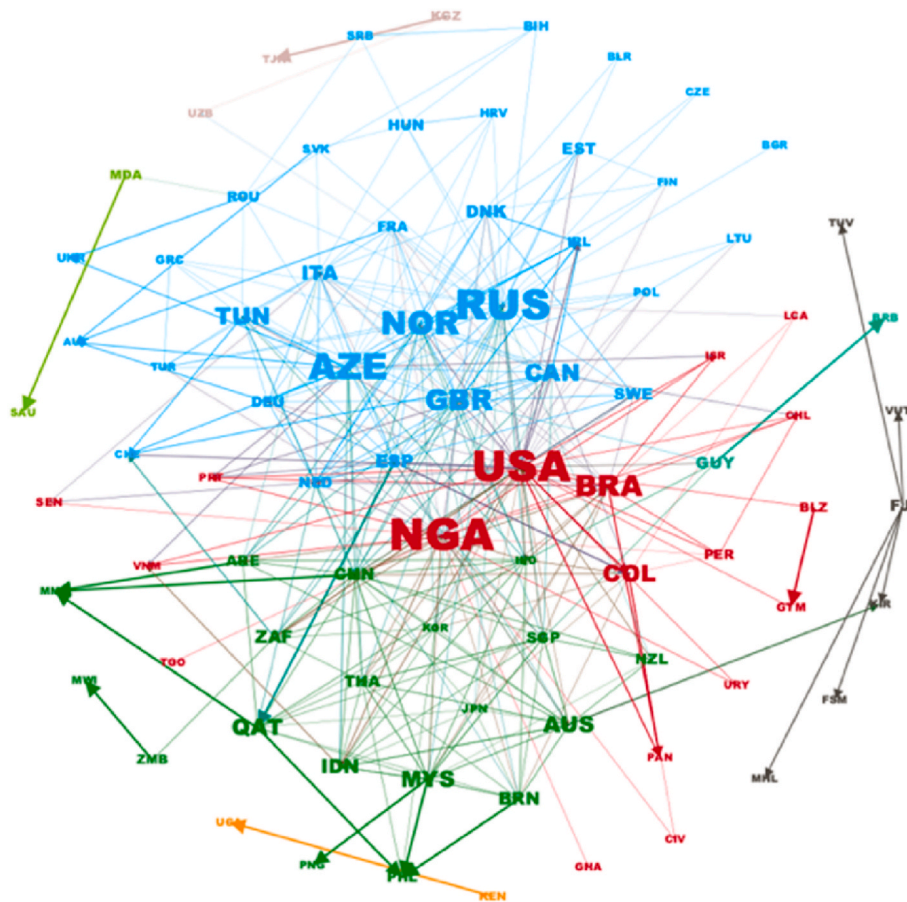


Fig. 4. Avalanche network in the Scenario of $r = 1.5$ in 2021. (Note: each point represents a country involved in oil trade; different colors indicate different modularity; Node and its label font size correspond to the value of out-degree. Larger node size signifies a greater avalanche size and wider influence. The width of an edge indicates the oil trade weight; the edge's direction denotes the oil trade flow).

Table 6

The structural dependence results of oil trade network on the impact of supply crisis in 2021 by ERGM. Normal oil trade shows the actual GOTN characteristics. Small and large oil supply crisis represent the avalanche network with parameter ratio $\alpha/\beta = 1$ and $\alpha/\beta = 1.5$.

Network Statistics	Normal oil trade	Small oil supply crisis	Large oil supply crisis
Edges	-0.505225 ^a (0.117)	-4.57325 ^a (0.152)	-4.83695 ^a (0.129)
Reciprocity	0.94921 ^a (0.276)	0.95174 ^b (0.296)	0.68700 ^c (0.240)
Cyclicity	-0.78474 ^a (0.091)	-0.49052 ^a (0.091)	-0.40964 ^a (0.084)
Transitivity	1.48839 ^a (0.120)	2.16871 ^a (0.0156)	2.34476 ^a (0.137)

Note: Standard deviation in parentheses. Signif. codes: 0.

^a 0.001.

^b 0.01.

^c 0.05 ‘,’ 0.1 ‘’ 1.

5. Discussion and conclusion

5.1. Main conclusion

The role of network dependencies is pivotal in comprehending the determinants of intricate trade dynamics evolution mechanisms. This study delves into the evolutionary mechanisms of structural dependence characteristics within the global oil trade network. Recognizing the effectiveness and adaptability of exponential random graph models (ERGMs) in revealing network dependency structures, we primarily employ temporal ERGM (TERGM) and separable TERGM (STERGM)

within the ERGM family to scrutinize the dynamic evolution of structural dependence within the oil trade network. Subsequently, we extend our analysis to explore the influence of supply crisis shocks on the network's structural dependence, utilizing a combination of the cascading failure model and the ERGM.

First, the United States and Russia have consistently held core positions within the global petroleum trade network. As China's oil trade relationships diversify, its centrality is gradually increasing. According to the endogenous attribute structural dependence characteristics analyses, our results suggest that the reciprocal trade structure has been crucial in establishing trade relations, particularly since the 2008 financial crisis. Global countries have increasingly emphasized the importance of reciprocal cooperation. Cyclicity and transitivity have witnessed significantly increased, signifying countries' inclination to explore new trade connections. Second, the influence of GDP, eigenvector centrality, betweenness centrality and modularity have shown a downward trend. This reflects the emergence of new characteristics in the world oil trade pattern. Third, external shocks lead to dramatic changes in the trade relationships between countries, concurrently altering the structural dependence features of the global oil network. As the scale of the supply crisis expands, risk spreads along the three core community networks led by the United States, Nigeria, and Russia, resulting in a significant reduction in structural dependence characteristics. Reciprocity's influence tends to diminish while transitivity's impact strengthens significantly. The susceptibility to external shocks and the contagion effect of cascading failures could propagate through trade links, even causing systemic collapse.

5.2. Policy implications

Based on the conclusion, we provide several policy implications as follows.

- (1) The oil trade network’s dependence structure is relatively fragile despite its substantial scale. Countries should continually bolster diversified trade cooperation and dependency relationships to withstand the impacts of global economic policy uncertainty. In particular, for economies heavily reliant on oil imports such as China, active participation in and promotion of multilateral trade negotiations are particularly crucial, which allows China to diversify its energy supply sources in the global oil market, reduce dependency on specific import origins, and mitigate risks associated with supply disruptions or price fluctuations.
- (2) Traditional indicators such as GDP and centrality are experiencing a decline in influence within the oil trade network, indicating a diversification of factors impacting the establishment of petroleum trade relationships. Factors such as geopolitical considerations, climate change risk, and new energy technologies are emerging as significant constraints on the oil trade. Policymakers should consider optimizing risk management by developing comprehensive risk assessment frameworks and establishing warming systems.
- (3) The trade position of countries within the oil trade network is primarily determined by differences in resource endowments, especially in recent years, as the Middle East region has rapidly increased its dominance in the oil trade. Policymakers should consider developing new energy technology to reduce dependence on petroleum resources. While China’s position in the global oil trade has improved, its ability to withstand shocks in the oil supply remains weak. China needs to focus on building a more resilient and robust supply chain system, and delicate efforts to develop clean energy and new energy technologies.

This research can be extended in the future. The oil trade can be segmented into supply, transport and output sections, and a multilayer oil trade network can be built. We can determine the structural dependence characteristics of the oil trade via the ERGMs. Alternatively, the petroleum industry can be subdivided to study the evolution mechanism of structural dependence among industries and regions.

CRediT authorship contribution statement

Jingjing Yan: Writing – original draft, Data curation, Methodology,

Appendix A.1

Table A.1
Country names and Three-Letter Codes table.

Country	Code	Country	Code	Country	Code
Aruba	ABW	Gibraltar	GIB	Nepal	NPL
Afghanistan	AFG	Guinea	GIN	Nauru	NRU
Angola	AGO	Guadeloupe	GLP	New Zealand	NZL
Albania	ALB	Gambia	GMB	Oman	OMN
Andorra	AND	Guinea-Bissau	GNB	Pakistan	PAK
Netherland Antilles	ANT	Equatorial Guinea	GNQ	Palestine	PAL
United Arab Emirates	ARE	Greece	GRC	Panama	PAN
Argentina	ARG	Grenada	GRD	Pitcairn	PCN
Armenia	ARM	Greenland	GRL	Peru	PER
French Southern Antartic territories	ATF	Guatemala	GTM	Philippines	PHL
Antigua and Barbuda	ATG	French Guiana	GUF	Palau	PLW
Australia	AUS	Guyana	GUY	Papua New Guinea	PNG

(continued on next page)

Software. **Yaoqi Guo**: Funding acquisition, Formal analysis, Project administration, Supervision, Writing – review & editing. **Hongwei Zhang**: Supervision, Writing – review & editing, Formal analysis, Project administration.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

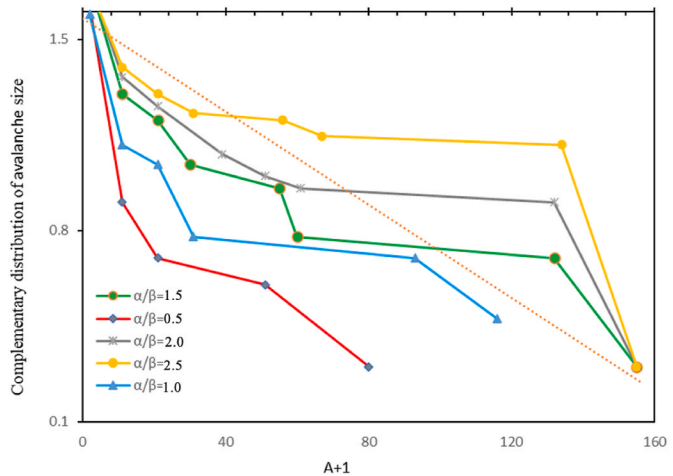


Fig. A2. Avalanche size distributions. Complementary distribution of avalanche size represents the distribution of avalanche sizes equal to or greater than A. We adjust the horizontal axis by one to plot the economies with A = 0 on a logarithmic scale [52].

Data availability

Data will be made available on request.

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Table A.1 (continued)

Country	Code	Country	Code	Country	Code
Austria	AUT	Honduras	HND	Poland	POL
Azerbaijan	AZE	Croatia	HRV	Puerto Rico	PRI
Burundi	BDI	Haiti	HTI	North Korea	PRK
Belgium and Luxembourg	BEL	Hungary	HUN	Portugal	PRT
Benin	BEN	Indonesia	IDN	Paraguay	PRY
Burkina Faso	BFA	India	IND	French Polynesia	PYF
Bangladesh	BGD	Ireland	IRL	Qatar	QAT
Bulgaria	BGR	Iran	IRN	Reunion	REU
Bahrain	BHR	Iraq	IRQ	Romania	ROM
Bahamas	BHS	Iceland	ISL	Russian Federation	RUS
Bosnia and Herzegovina	BIH	Israel	ISR	Rwanda	RWA
Belarus	BLR	Italy	ITA	Saudi Arabia	SAU
Belize	BLZ	Jamaica	JAM	Sudan	SDN
Bermuda	BMU	Jordan	JOR	Senegal	SEN
Bolivia	BOL	Japan	JPN	Singapore	SGP
Brazil	BRA	Kazakstan	KAZ	Saint Helena	SHN
Barbados	BRB	Kenya	KEN	Solomon Islands	SLB
Brunei Darussalam	BRN	Kyrgyzstan	KGZ	Sierra Leone	SLE
Bhutan	BTN	Cambodia	KHM	El Salvador	SLV
Botswana	BWA	Kiribati	KIR	San Marino	SMR
Central African Republic	CAF	Saint Kitts and Nevis	KNA	Somalia	SOM
Canada	CAN	Korea	KOR	St. Pierre and Miquelon	SPM
Cocos (Keeling) Islands	CCK	Kuwait	KWT	Sao Tome and Principe	STP
Switzerland	CHE	Lao People's Democratic Republic	LAO	Suriname	SUR
Chile	CHL	Lebanon	LBN	Slovakia	SVK
China	CHN	Liberia	LBR	Slovenia	SVN
Country	Code	Country	Code	Country	Code
Côte d'Ivoire	CIV	Libyan Arab Jamahiriya	LBY	Sweden	SWE
Cameroon	CMR	Saint Lucia	LCA	Swaziland	SWZ
Congo	COG	Sri Lanka	LKA	Seychelles	SYC
Cook Islands	COK	Lesotho	LSO	Syrian Arab Republic	SYR
Colombia	COL	Lithuania	LTU	Turks and Caicos Islands	TCA
Comoros	COM	Luxembourg	LUX	Chad	TCO
Cape Verde	CPV	Latvia	LVA	Togo	TGO
Costa Rica	CRI	Macao (Aomen)	MAC	Thailand	THA
Cuba	CUB	Morocco	MAR	Tajikistan	TJK
Christmas Island	CXR	Moldova	MDA	Tokelau	TKL
Cayman Islands	CYM	Madagascar	MDG	Turkmenistan	TKM
Cyprus	CYP	Maldives	MDV	East Timor	TMP
Czech Republic	CZE	Mexico	MEX	Tonga	TON
Germany	DEU	Marshall Islands	MHL	Trinidad and Tobago	TTO
Djibouti	DJI	Macedonia	MKD	Tunisia	TUN
Dominica	DMA	Mali	MLI	Turkey	TUR
Denmark	DNK	Malta	MLT	Tuvalu	TUV
Dominican Republic	DOM	Burma	MMR	Tanzania, United Rep. of	TZA
Algeria	DZA	Mongolia	MNG	Uganda	UGA
Ecuador	ECU	Northern Mariana Islands	MNP	Ukraine	UKR
Egypt	EGY	Mozambique	MOZ	Uruguay	URY
Eritrea	ERI	Mauritania	MRT	United States of America	USA
Western Sahara	ESH	Montserrat	MSR	Uzbekistan	UZB
Spain	ESP	Martinique	MTQ	Saint Vincent and the Grenadines	VCT
Estonia	EST	Mauritius	MUS	Venezuela	VEN
Ethiopia	ETH	Malawi	MWI	British Virgin Islands	VGB
Finland	FIN	Malaysia	MYS	Viet Nam	VNM
Fiji	FJI	Namibia	NAM	Vanuatu	VUT
Falkland Islands	FLK	New Caledonia	NCL	Wallis and Futuna	WLF
France	FRA	Niger	NER	Samoa	WSM
Faroe Islands	FRO	Norfolk Island	NFK	Yemen	YEM
Micronesia (Federated States of)	FSM	Nigeria	NGA	Serbia and Montenegro	YUG
Gabon	GAB	Nicaragua	NIC	South Africa	ZAF
United Kingdom	GBR	Niue	NIU	Congo	ZAR
Georgia	GEO	Netherlands	NLD	Zambia	ZMB
Ghana	GHA	Norway	NOR	Zimbabwe	ZWE

Note: the codes are issued by the National Standardization Management Committee, with the standard number GB/T 2658-2000.

Appendix A.2

Following the extant literature [52,54,55], the dynamic of supply crisis propagation across the network depends on the α/β ratio. Depending on the value of α/β , the avalanche size distribution $P(A)$ has a qualitatively different forms. When α/β is too large, there is an excess global avalanches, leading to a finite peak around the range of avalanche sizes in the network. When the α/β is too small, there is no large avalanches, resulting in a rapid decay of $P(A)$ with A . Between these extremes, there exists a critical point. Thus, we plot the complementary distribution of avalanche size in log scale to capture the tail characteristics. From the graph Fig. A1, we can see that the distributions of avalanche size before and after $\alpha/\beta = 1.5$ are strikingly different, where $P(A)$ follows a power-law distribution and exhibit the most widespread avalanche outcomes in the system. When $\alpha/\beta < 1.5$, the

largest avalanche size reaches to 80 (50 % of the economies are affected). Whereas when $\alpha/\beta > 1.5$, the largest avalanche size increases from 80 to 154 (92 % of the economies are affected). Therefore, $\alpha/\beta = 1.5$ is taken as a key value of model parameter corresponding to the scenario analysis in the next section.

References

- [1] Li Y, Chen B, Chen G, Wu X. The global oil supply chain: the essential role of non-oil product as revealed by a comparison between physical and virtual oil trade patterns. *Resour Conserv Recycl* 2021;175:105836.
- [2] Dudley B. *Bp statistical review of world energy*. London, UK: BP Statistical Review; 2018. 2019).
- [3] Wang M, Tian L, Du R. Research on the interaction patterns among the global crude oil import dependency countries: a complex network approach. *Appl Energy* 2016; 180:779–91.
- [4] Watts DJ, Strogatz SH. Collective dynamics of ‘small-world’ networks. *Nature* 1998;393(6684):440–2.
- [5] Zhang YJ, Wang JL. Do high-frequency stock market data help forecast crude oil prices? Evidence from the MIDAS models. *Energy Econ* 2019;78:192–201.
- [6] De Benedictis L, Tajoli L. The world trade network. *World Econ* 2011;34(8): 1417–54.
- [7] Deguchi T, Takahashi K, Takayasu H, Takayasu M. Hubs and authorities in the world trade network using a weighted HITS algorithm. *PLoS One* 2014;9(7): e100338.
- [8] An H, Zhong W, Chen Y, Li H, Gao X. Features and evolution of international crude oil trade relationships: a trading-based network analysis. *Energy* 2014;74:254–9.
- [9] Fracasso A, Nguyen HT, Schiavo S. The evolution of oil trade: a complex network approach. *Network Science* 2018;6(4):545–70.
- [10] Gao C, Sun M, Shen B. Features and evolution of international fossil energy trade relationships: a weighted multilayer network analysis. *Appl Energy* 2015;156: 542–54.
- [11] Ji Q, Zhang HY, Fan Y. Identification of global oil trade patterns: an empirical research based on complex network theory. *Energy Convers Manag* 2014;85: 856–65.
- [12] Du R, Wang Y, Dong G, Tian L, Liu Y, Wang M, Fang G. A complex network perspective on interrelations and evolution features of international oil trade, 2002–2013. *Appl Energy* 2017;196:142–51.
- [13] Wei N, Xie WJ, Zhou WX. Robustness of the international oil trade network under targeted attacks to economies. *Energy* 2022;251:123939.
- [14] Xi X, Zhou J, Gao X, Liu D, Zheng H, Sun Q. Impact of changes in crude oil trade network patterns on national economy. *Energy Econ* 2019;84:104490.
- [15] Bechtel W, Abrahamsen A. Explanation: a mechanist alternative. *Stud Hist Philos Sci C Stud Hist Philos Biol Biomed Sci* 2005;36(2):421–41.
- [16] Krivitsky PN, Handcock MS. A separable model for dynamic networks. *J Roy Stat Soc B Stat Methodol* 2014;76(1):29–46.
- [17] Craver C, Tabery J. Mechanisms in science. In: Zalta EN, editor. *The Stanford Encyclopedia of Philosophy*; 2015.
- [18] Herman PR. Modeling complex network patterns in international trade. *Rev World Econ* 2022;158(1):127–79.
- [19] Pan Z. Varieties of intergovernmental organization memberships and structural effects in the world trade network. *Adv Complex Syst* 2018;21(2):1850001.
- [20] Hanneke S, Fu W, Xing EP. Discrete temporal models of social networks. *Electronic Journal of Statistics* 2010;4:585–605.
- [21] Ward MD, Ahlquist JS, Rozenas A. Gravity’s rainbow: a dynamic latent space model for the world trade network. *Network Science* 2013;1(1):95–118.
- [22] Chen B, Xu H, Feng L. Structural properties and evolution of global photovoltaic industry trade network. *Environ Sci Pollut Control Ser* 2023;30(26):69580–98.
- [23] Gao C, Tao S, Su B, Mensah IA, Sun M. Exploring renewable energy trade cooperation relationships: Evidence from belt and road countries, 1996–2018. *Renew Energy* 2023;202:196–209.
- [24] Kitamura T, Managi S. Driving force and resistance: network feature in oil trade. *Appl Energy* 2017;208:361–75.
- [25] Yang X, Xiao F. An improved gravity model to identify influential nodes in complex networks based on k-shell method. *Knowl Base Syst* 2021;227:107198.
- [26] Zhao J, Wang Y, Deng Y. Identifying influential nodes in complex networks from global perspective. *Chaos, Solit Fractals* 2020;133:109637.
- [27] Du R, Dong G, Tian L, Wang Y, Liu Y, Wang M, Fang G. A complex network perspective on features and evolution of world crude oil trade. *Energy Proc* 2016; 104:221–6.
- [28] Zhang HY, Ji Q, Fan Y. Competition, transmission and pattern evolution: a network analysis of global oil trade. *Energy Pol* 2014;73:312–22.
- [29] Chen Z, An H, An F, Guan Q, Hao X. Structural risk evaluation of global gas trade by a network-based dynamics simulation model. *Energy* 2018;159:457–71.
- [30] Hao X. Import competition and pressure in the international crude oil trade: a network analysis. *Resour Pol* 2023;82:103468.
- [31] Wang W, Fan LW, Zhou P. Evolution of global fossil fuel trade dependencies. *Energy* 2022;238:121924.
- [32] Chen Z, Wang H, Liu X, Wang Z, Wen S. Risk diffusion of international oil trade cuts: a network-based dynamics model. *Energy Rep* 2022;8:11320–33.
- [33] Sun Q, Gao X, Zhong W, Liu N. The stability of the international oil trade network from short-term and long-term perspectives. *Phys Stat Mech Appl* 2017;482: 345–56.
- [34] Xi X, Zhou J, Gao X, Wang Z, Si J. Impact of the global mineral trade structure on national economies based on complex network and panel quantile regression analyses. *Resour Conserv Recycl* 2020;154:104637.
- [35] Xia Q, Du D, Yu Z, Li X, Zhang Q. Coins have both sides: Revealing the structure and pattern of global interdependence network for five critical metals. *Resources Policy* 2024;88:104453.
- [36] Zhang W, Chong Z, Li X, Nie G. Spatial patterns and determinant factors of population flow networks in China: analysis on Tencent Location Big Data. *Cities* 2020;99:102640.
- [37] Kang X, Wang M, Chen L, Li X. Supply risk propagation of global copper industry chain based on multi-layer complex network. *Resour Pol* 2023;85:103797.
- [38] Leng Z, Shuai J, Sun H, Shi Z, Wang Z. Do China’s wind energy products have potentials for trade with the “Belt and Road” countries?—A gravity model approach. *Energy Pol* 2020;137:111172.
- [39] Zhang C, Fu J, Pu Z. A study of the petroleum trade network of countries along “The Belt and Road Initiative”. *J Clean Prod* 2019;222:593–605.
- [40] Chen W, Niu X, Ke W, Yu Z. Investigating the energy trade networks in the belt and road regions: structures and evolution. *Energy* 2023;283:129157.
- [41] Harris JK. *An introduction to exponential random graph modeling*, vol. 173. Sage Publications; 2013.
- [42] Leifeld P, Cranmer SJ, Desmarais BA. Temporal exponential random graph models with btergm: estimation and bootstrap confidence intervals. *J Stat Software* 2018; 83(6).
- [43] Krivitsky PN, Koehly LM, Marcum CS. Exponential-family random graph models for multi-layer networks. *Psychometrika* 2020;85(3):630–59.
- [44] Kim JY, Howard M, Cox Pahnke E, Boeker W. Understanding network formation in strategy research: exponential random graph models. *Strat Manag J* 2016;37(1): 22–44.
- [45] Contractor NS, Wasserman S, Faust K. Testing multitheoretical, multilevel hypotheses about organizational networks: an analytic framework and empirical example. *Acad Manag Rev* 2006;31(3):681–703.
- [46] Gao J, Yu X. Factors affecting the evolution of technical cooperation among “belt and road initiative” countries based on TERGMs and ERGMs. *Sustainability* 2022; 14(3):1760.
- [47] Lee KH, Xue L, Hunter DR. Model-based clustering of time-evolving networks through temporal exponential-family random graph models. *J Multivariate Anal* 2020;175:104540.
- [48] Matous P, Wang P, Lau L. Who benefits from network intervention programs? TERGM analysis across ten Philippine low-income communities. *Soc Network* 2021;65:110–23.
- [49] Cai H, Wang Z, Zhu Y. Understanding the structure and determinants of intercity carbon emissions association network in China. *J Clean Prod* 2022;352:131535.
- [50] Guo Y, Zhao B, Zhang H. The impact of the Belt and Road Initiative on the natural gas trade: a network structure dependence perspective. *Energy* 2023;263:125912.
- [51] Wang XY, Chen B, Hou N, Chi ZP. Evolution of structural properties of the global strategic emerging industries’ trade network and its determinants: An TERGM analysis. *Industrial Marketing Management* 2024;118:78–92.
- [52] Lee KM, Yang JS, Kim G, Lee J, Goh KI, Kim IM. Impact of the topology of global macroeconomic network on the spreading of economic crises. *PLoS One* 2011;6(3): e18443.
- [53] Kang H, Lee KM, Yang JS. The potential for cascading failures in the international trade network. *PLoS One* 2024;19(3):e0299833.
- [54] Sun X, Shi Q, Hao X. Supply crisis propagation in the global cobalt trade network. *Resour Conserv Recycl* 2022;179:106035.
- [55] Wang X, Li H, Yao H, Zhu D, Liu N. Simulation analysis of the spread of a supply crisis based on the global natural graphite trade network. *Resour Pol* 2018;59: 200–9.
- [56] Lusher D, Koskinen J, Robins G, editors. *Exponential random graph models for social networks: theory, methods, and applications*. Cambridge University Press; 2013.
- [57] Hunter DR, Handcock MS. Inference in curved exponential family models for networks. *J Comput Graph Stat* 2006;15(3):565–83.